

Overview of Image to image translation

Subject: Image to image translation

1.

Graph partitioning methods Graph partitioning methods are an effective tools for image segmentation since they model the impact of pixel neighborhoods on a given cluster of pixels or pixel, under the assumption of homogeneity in images. In these methods, the image is modeled as a weighted, undirected graph. Usually a pixel or a group of pixels are associated with nodes and edge weights define the (dis)similarity between the neighborhood pixels. The graph (image) is then partitioned according to a criterion designed to model "good" clusters. Each partition of the nodes (pixels) output from these algorithms are considered an object segment in the image; see Segmentation-based object categorization. Some popular algorithms of this category are normalized cuts, random walker, minimum cut, isoperimetric partitioning, minimum spanning tree-based segmentation, and segmentation-based object categorization.

2.

$$\alpha (1 - \delta(\mathbf{I} - \mathbf{I}_{\text{initial}})) + \beta \sum_{q \in N(i)} (1 - \delta(\mathbf{I}_i, \mathbf{I}_q)) . \quad \{\displaystyle \alpha (1 - \delta(\ell_i - \ell_{\text{initial}})) + \beta \sum_{q \in N(i)} (1 - \delta(\ell_i, \ell_{q(i)}))\} .$$

3.

The functional value is the sum of the total length of the segmentation curve K $\{\displaystyle K\}$, the smoothness of the approximation u $\{\displaystyle u\}$, and its distance to the original image f $\{\displaystyle f\}$. The weight of the smoothness penalty is adjusted by $\mu > 0$ $\{\displaystyle \mu > 0\}$. The Potts model is often called piecewise constant Mumford-Shah model as it can be seen as the degenerate case $\mu \rightarrow \infty$ $\{\displaystyle \mu \rightarrow \infty\}$. The optimization problems are known to be NP-hard in general but near-minimizing strategies work well in practice. Classical algorithms are graduated non-convexity and Ambrosio-Tortorelli approximation.

4.

Here $\lambda \in \Lambda$ $\{\displaystyle \lambda \in \Lambda\}$, the set of all possible labels. 3. M step: The established relevance of a given feature set to a labeling scheme is now used to compute the a priori estimate of a given label in the second part of the algorithm. Since the actual number of total labels is unknown (from a training data set), a hidden estimate of the number of labels given by the user is utilized in computations.

5.

Model-based segmentation The central assumption of model-based approaches is that the structures of interest have a tendency towards a particular shape. Therefore, one can seek a probabilistic model that characterizes the shape and its variation. When segmenting an image, constraints can be imposed using this model as a prior. Such a task may involve (i) registration of the training examples to a common pose, (ii) probabilistic representation of the variation of the registered samples, and (iii) statistical inference between the model and the image. Other important methods in the literature for model-based segmentation include active shape models and active appearance models.

6.

This mathematical expression can be implemented by convolving with an appropriate mask. If we extend this equation to three dimensions (x,y,z), the intensity at each pixel location around a central pixel at (x, y, z) is replaced by their corresponding values. This equation becomes particularly useful when we assume that all pixels have unit spacing along each axis. A sphere mask has been developed for use with three-dimensional datasets. The sphere mask is designed to use only integer arithmetic during calculations, thereby eliminating the need for floating-point hardware or software. When applying these concepts to actual images represented as arrays of numbers, we need to consider what happens when we reach an edge or border region. The function $g(x, y)$ is defined as:

7.

Optimization algorithms Each optimization algorithm is an adaptation of models from a variety of fields and they are set apart by their unique cost functions. The common trait of cost functions is to penalize change in pixel value as well as difference in pixel label when compared to labels of neighboring pixels.

8.

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

9.

Content-based image retrieval Machine vision Medical imaging, and imaging studies in biomedical research, including volume rendered images from computed tomography, magnetic resonance imaging, as well as volume electron microscopy techniques such as FIB-SEM. Locate tumors and other pathologies Measure tissue volumes Diagnosis, study of anatomical structure Surgery planning Virtual surgery simulation Intra-surgery navigation Radiotherapy Digital Pathology and Histopathology. Nuclei instance segmentation in variously stained whole slide images (WSIs) refers to the automatic delineation of individual cells' nuclei borders. This task is a specialized subproblem of instance segmentation, with high practical importance in biomedical research and diagnostics (e.g. cell counting, morphometric feature extraction, tumor grading, prognostic biomarker extraction). The modern approach to the problem is the use of deep learning architectures developed to encompass the main challenges of separating overlapping or fused nuclei. The availability of few humanly-annotated datasets poses another challenge to the problem. Object detection Pedestrian detection Face detection Brake light detection Locate objects in satellite images (roads, forests, crops, etc.) Recognition Tasks Face recognition Fingerprint recognition Iris recognition Prohibited Item at Airport security checkpoints Traffic control systems Video surveillance Video object co-segmentation and action localization Several general-purpose algorithms and techniques have been developed for image segmentation. To be useful, these techniques must typically be combined with a domain's specific knowledge in order to effectively solve the domain's segmentation problems.

10.

Variational methods The goal of variational methods is to find a segmentation which is optimal with respect to a specific energy functional. The functionals consist of a data fitting term and a regularizing

terms. A classical representative is the Potts model defined for an image f by

11.

Disadvantages of MAP and EM based image segmentation Exact MAP estimates cannot be easily computed. Approximate MAP estimates are computationally expensive to calculate. Extension to multi-class labeling degrades performance and increases storage required. Reliable estimation of parameters for EM is required for global optima to be achieved. Based on method of optimization, segmentation may cluster to local minima.